Numerical Methods: Simpson's Rule

Questions 1–21

Question 1. Using Simpson's rule, calculate an approximation to the integral

$$\int_0^2 e^{-x^3}\,dx$$

for each of the following, giving your answer to 6 decimal places:

a) Three ordinates.
b) Five ordinates.
c) Seven ordinates.

Question 2. Using Simpson's rule, calculate an approximation to the integral

$$\int_1^2 \frac{1}{1+\ln x}\,dx$$

for each of the following, giving your answer to 6 decimal places:

a) Two strips.
b) Four strips.
c) Six strips.

Question 3. The table below shows the velocity $v\,\mathrm{m\,s^{-1}}$ of a car every 2 seconds.

t	0	2	4	6	8
v	0	5.6	9.8	13.0	15.2

Use Simpson's rule to estimate the distance travelled in the first 8 seconds.

Prerequisite knowledge for Q4: Hyperbolic Functions

Question 4. Use Simpson's rule with five ordinates to find an estimate for

$$\int_0^1 \cosh x \, dx,$$

giving your answer to three significant figures.

Question 5.

a) Use Simpson's rule with four strips of equal width to find an estimate for

$$\int_{0.1}^{0.5} \frac{\tan x}{x}\, dx,$$

giving your answer to 3 significant figures.

b) Explain how you could find an improved estimate.

Question 6.

a) Use Simpson's rule with three ordinates to estimate

$$\int_{0}^{\pi} e^{\cos x}\, dx,$$

giving your answer to four decimal places.

b) Explain how Simpson's rule could be used to find a more accurate estimate of the integral in part (a).

Question 7.

a) Use Simpson's rule with four strips to find an estimate for

$$\int_0^1 e^x \, dx,$$

giving your answer to five significant figures.

b) Find the exact value of $\int_0^1 e^x \, dx$.

c) Find the relative error of your approximation from part (a).

Prerequisite knowledge for Q8: Trapezium Rule

Question 8. It is given that

$$I = \int_0^1 (x^2 + e^{2x}) \, dx.$$

a) Find an estimate for I using the trapezium rule with four strips, giving your answer to four decimal places.

b) Find an estimate for I using Simpson's rule with four strips, giving your answer to four decimal places.

c) Find the exact value of I.

Prerequisite knowledge for Q9: Reversing the Chain Rule or Integration by Substitution

Question 9.

a) Use Simpson's rule with two strips to find an estimate for

$$\int_0^2 x\sqrt{2x^2+1}\,dx,$$

giving your answer to four decimal places.

b) Find the exact value of $\displaystyle\int_0^2 x\sqrt{2x^2+1}\,dx$.

c) Find the relative error of your approximation from part (a).

Question 10. A curve has equation $y = \cos(x^2)$.

a) Use Simpson's rule with four strips to find an estimate for $\displaystyle\int_0^1 \cos(x^2)\,dx$, giving your answer to four decimal places.

b) The graph with equation $y = \cos(x^2)$ is stretched with scale factor 5 parallel to the y-axis and then translated by 1 parallel to the x-axis to give the graph with equation $y = f(x)$. Write down an expression for $f(x)$.

c) Giving your answers to three decimal places, using your answer to part (a), state an estimate for:

i) $\displaystyle\int_1^2 f(x)\,dx$.

d) Use Simpson's rule with four strips to find an estimate for $\displaystyle\int_1^2 f(x)\,dx$.

Question 11.

a) Use Simpson's rule with five ordinates to find an approximation to

$$\int_{0.1}^{0.3} \frac{1}{\sqrt{1-x^3}}\,dx,$$

giving your answer to three decimal places.

b) Hence find an approximation to $\displaystyle\int_{0.1}^{0.3} \frac{5}{\sqrt{1-x^3}}\,dx$.

Question 12.

a) Use Simpson's rule with five ordinates to find an approximation to

$$\int_{1}^{3} \frac{\sin x}{x}\,dx,$$

giving your answer to three decimal places.

b) Hence find an approximation to $\displaystyle\int_{1}^{3} \left(3 - \frac{\sin x}{x}\right)\,dx$.

Prerequisite knowledge for Q13: Volume of Revolution

Question 13. A curve has equation $y = x\ln(x+2)$. The region bounded by the curve, the lines $x = 0$ and $x = 2$, and the x-axis is rotated 360° about the x-axis.

a) Write down an expression for the exact volume of revolution.

b) Use Simpson's rule with five ordinates to estimate this volume, giving your answer to 3 significant figures.

Prerequisite knowledge for Q14: Mean Value

Question 14.

a) Sketch the curve C with equation $y = 1 + \sin(\frac{x}{2})$ for $0 \leq x \leq 2\pi$.

b) The region bounded by the curve C, the coordinate axes, and the vertical straight line with equation $x = 2\pi$ is to be estimated by Simpson's rule using six equally spaced strips. Find, correct to six decimal places, an approximation to the area of this region.

c) Hence estimate the mean value of $y = 1 + \sin(\frac{x}{2})$ in the interval $0 \leq x \leq 2\pi$, giving your answer to five decimal places.

Question 15. Given that $\ln a = \int_1^a \frac{1}{x}\,dx$, use Simpson's rule with seven ordinates to find an estimate for $\ln 7$, giving your answer to four decimal places.

Prerequisite knowledge for Q16–Q17: Maclaurin Series

Question 16.

a) Find the first three non-zero terms in the Maclaurin expansion of $\sin 2x$.

b) Use Simpson's rule with four strips and this expansion to estimate $\int_0^1 \sin 2x\,dx$, giving your answer to four decimal places.

Question 17.

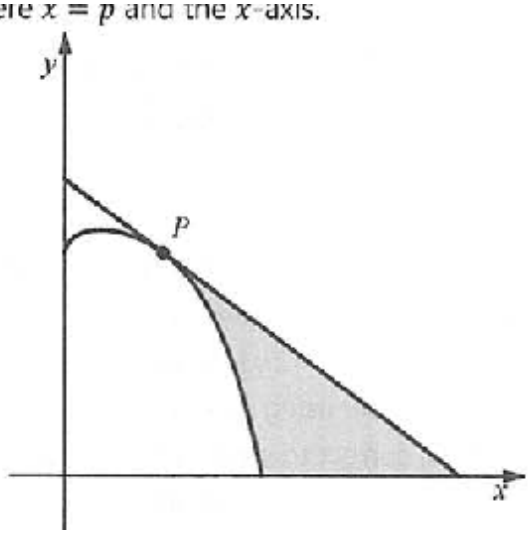
a) Find the first three non-zero terms in the expansion of $x^2 \ln(1+x)$.

b) Use Simpson's rule with four strips and this expansion to estimate $\int_0^{0.4} x^2 \ln(1+x)\,dx$, giving your answer to four decimal places.

Prerequisite knowledge for Q18: The Chain Rule

Question 18. The shaded regions on the diagram are bounded by the curve with equation $y = -x^x + 4$ for $1 \leq x \leq 2$, the tangent line at point P where $x = p$, and the x-axis. It is given that $p = 1$.

where $x = p$ and the x-axis.



a) State the coordinates of the point P.

b) Use Simpson's rule with four strips to find an estimate for $\int_1^2 (-x^x + 4)\,dx$, giving your answer to two decimal places.

c) Given that $x^x = e^{x \ln x}$, show that $\dfrac{d}{dx}(x^x) = x^x(1 + \ln x)$.

d) Find the exact form of the equation of the tangent line, giving your answer in the form $ax + by = c$, where a, b and c are integers.

e) Hence find an approximate value for the total area of the shaded regions, giving your answer to two decimal places.

Prerequisite knowledge for Q19–Q21: Trapezium Rule, Mid-ordinate Rule and Reversing the Chain Rule or Integration by Substitution

Question 19. Let $I = \int_0^2 x^2 \, dx$.

a) Find an estimate for I, giving your answer as accurately as possible, using:
   i) the trapezium rule with four strips, T_4;
   ii) the mid-ordinate rule with four strips, M_4;
   iii) Simpson's rule with eight strips, S_8.

b) Find the exact value of I.

c) Show that $\frac{2M_4 + T_4}{3} = S_8$.

Question 20. Let $I = \int_0^6 \sqrt{x+1} \, dx$.

a) Find an estimate for I, giving your answer to six decimal places, using:
   i) the trapezium rule with four strips, T_4;
   ii) the mid-ordinate rule with four strips, M_4;
   iii) Simpson's rule with eight strips, S_8.

b) Find the exact value of I.

c) Show that $\frac{2M_4 + T_4}{3} = S_8$.

Question 21. Let $I = \int_a^b x^2\,dx$, where a and b are positive constants such that $a < b$. Prove each of the following:

a) The trapezium rule with two strips, T_2, is given by $\frac{1}{8}(b-a)(3a^2+2ab+3b^2)$.

b) The mid-ordinate rule with two strips, M_2, is given by $\frac{1}{16}(b-a)(5a^2+6ab+5b^2)$.

c) Simpson's rule with four strips, S_4, is given by $\frac{1}{3}(b-a)(a^2+ab+b^2)$.

d) The exact value of I is given by $\frac{1}{3}(b-a)(a^2+ab+b^2)$.

e) $\frac{2M_2+T_2}{3} = S_4$.